A Reproducible Pipeline for Landslide Segmentation Dataset Development in Nepal

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***Abstract*—This article presents a reproducible multi-source workflow that converts reported landslide incidents in Nepal into reviewed data for segmentation-dataset development. The workflow operates in three stages. First, incident dates and geographic zones from the Building Information Platform Against Disaster (BIPAD) guide the acquisition of pre-event and post-event Planet optical mosaics, Sentinel-1 synthetic aperture radar (SAR), Sentinel-2 support imagery, and Shuttle Radar Topography Mission (SRTM) terrain observations. Second, optical and SAR changes are estimated independently using Iteratively Reweighted Multivariate Alteration Detection (IR-MAD), fused through a geometric mean, constrained by slope, and aggregated with Simple Linear Iterative Clustering (SLIC) superpixels. Segment-level chi-square evidence, Bonferroni-corrected significance, hysteresis thresholding, morphology, and geometric filtering are then used to extract and rank candidate regions. Third, candidates are reviewed through a QGIS (formerly Quantum GIS) plugin, and confirmed landslides are annotated for segmentation. The Hugging Face repository `sasudo2/landslides` provides centralized storage and provenance. The inventory currently contains 4,138 incidents from 2018–2026; candidate detection is complete for the first 730 incidents, with 248 landslide candidates and 5,595 non-landslide candidates reviewed. Pixel-level segmentation annotations have been completed for 164 landslides.**

***Index Terms*—landslide segmentation, dataset development, Nepal, Planet mosaics, IR-MAD, synthetic aperture radar, SLIC, change detection, manual annotation.**

I. Introduction

Nepal's steep terrain, active tectonics, monsoon rainfall, and expanding infrastructure produce frequent landslides, yet reported incidents rarely include boundaries suitable for training segmentation models. The Building Information Platform Against Disaster (BIPAD) provides dates and approximate locations, but converting these event records into reliable image–mask pairs still requires temporally appropriate imagery, spatial co-registration, candidate localization, and expert verification [17]. Manual satellite mapping can produce accurate inventories, as demonstrated after the 2015 Nepal earthquakes, but it remains costly and difficult to scale across many incidents [1].

Deep learning has increased the speed and automation of landslide mapping, but its performance remains strongly dependent on the quantity, diversity, and representativeness of training samples. A recent comprehensive review identifies sample dependence, limited generalization, and the continuing need for expert knowledge as major barriers to operational deep-learning-based landslide mapping [10]. The central problem is therefore not only the choice of segmentation architecture; it is the absence of a reproducible mechanism for transforming newly reported Nepalese incidents into traceable optical and synthetic aperture radar (SAR) candidate data that can be inspected and annotated.

II. Related Work and Contribution

Existing inventories and segmentation datasets provide important reference data, including the Nepal semantic segmentation dataset [3], the Asia Summer Monsoon inventory [4], and broader reviews of Nepalese landslide inventories [5]. More recent work has improved geographic and sensor diversity. For example, the Globally Distributed Coseismic Landslide Dataset integrates high-resolution imagery from several platforms across nine earthquake events and demonstrates its value for segmentation-model evaluation [11]. These resources advance benchmark development, but they do not fully solve the present use case.

Most existing datasets are distributed as fixed collections of images and completed labels. They do not provide an operational process for linking new BIPAD reports to incident-specific imagery, preserving acquisition and processing provenance, or extending the dataset as new incidents are added. Event-based datasets are also commonly centered on selected earthquake sequences, whereas this Nepal workflow accommodates heterogeneous reported incidents and uncertain coordinates. Furthermore, a completed segmentation dataset does

not provide the intermediate optical, SAR, terrain, and change-confidence evidence needed to discover previously unlabeled candidate locations and retain difficult non-landslide changes for later review.

The contribution of this work is therefore a reproducible dataset-generation workflow rather than another static benchmark or a new segmentation network. It connects BIPAD incidents to Planet and Sentinel-2 optical imagery, Sentinel-1 SAR, and Shuttle Radar Topography Mission (SRTM) observations; performs modality-specific unsupervised Iteratively Reweighted Multivariate Alteration Detection (IR-MAD) change analysis; applies statistically corrected segment selection and terrain constraints; and converts ranked candidates into human-reviewed labels and segmentation annotations. Its novelty lies in combining incident-driven acquisition, optical–SAR change fusion, family-wise significance control, rerun-safe provenance, candidate-level review, and segmentation-dataset construction within one continuously extensible system.

III. Methodology

The methodology is organized into three functional stages: image collection, statistically constrained candidate detection, and manual review with segmentation annotation. Incident information guides the acquisition of complementary optical, SAR, and terrain observations. These observations are aligned and analyzed for change before candidate regions are extracted for expert review. Confirmed landslide candidates are subsequently annotated at pixel level to support segmentation-dataset development. The Hugging Face dataset repository `sasudo2/landslides` provides centralized storage and provenance for the collected data, generated candidates, and published labels [7]. Fig. 1 summarizes the process.

BIPAD incidents → multi-source acquisition → statistically filtered candidates → manual review → segmentation annotations

Fig. 1. High-level workflow from reported incidents to a labeled landslide dataset.

A. Image Collection

Reported landslide locations and dates from 2018–2026 were collected from BIPAD. Each incident record defines a zone that bounds the search area but is not treated as the landslide location itself. Large zones are constrained around their centroid so that imagery acquisition and change analysis remain computationally manageable while preserving the reported incident context.

Pre-event and post-event optical observations are obtained from Planet quarterly mosaics. Mosaics intersecting an incident zone are combined and clipped to the area of interest. Sentinel-1 scenes immediately surrounding the incident provide pre-event and post-event vertical-transmit/vertical-receive (VV) and vertical-transmit/horizontal-receive (VH) backscatter observations. Google Earth Engine (GEE) is used to acquire and process Sentinel support data, derive slope and aspect from the SRTM digital elevation model, and obtain a cloud-masked Sentinel-2 pre-event fallback when required [22].

All acquired observations are retained with their incident identifiers in centralized storage. Processing is idempotent: completed assets are detected before acquisition, while failures in individual support layers do not invalidate otherwise usable incident data. Table I documents the spatial and temporal characteristics and acquisition window of every source dataset. Representative optical, SAR, slope, and aspect observations are shown in Fig. 2.

B. Candidate Detection

Candidate detection locates spatially coherent regions whose pre-event and post-event observations differ strongly enough to represent possible landslides. A detected candidate is not a confirmed landslide; it is a geographically bounded region prioritized for manual inspection. Change is first measured independently in optical imagery, which records surface reflectance, and SAR imagery, which records microwave backscatter. The resulting evidence is then combined with terrain and spatial-context constraints.

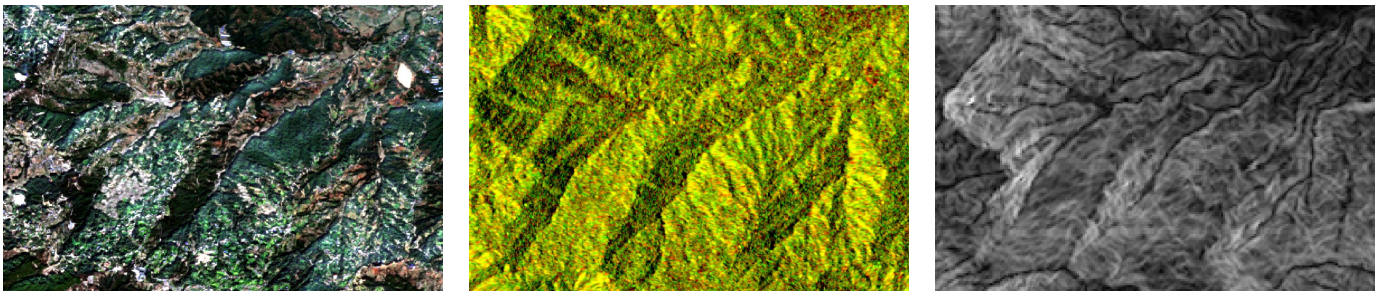
a) Spatial alignment: Spatial alignment, also called raster co-registration, places all observations on the same coordinate system, pixel spacing, extent, and pixel origin. This step is essential because change detection is meaningful only when corresponding pixels represent the same ground location [12]. Each optical, SAR, and terrain raster is reprojected and resampled onto a common analysis grid. The valid analysis area is then restricted to locations supported by every required observation, preventing missing or non-overlapping pixels from influencing the comparison.

b) IR-MAD transformation: IR-MAD is an unsupervised multivariate change-detection method derived from canonical correlation analysis [8]. It constructs linear combinations of the before- and after-event bands that are maximally correlated and then measures their differences. Unchanged pixels are expected to remain strongly correlated, whereas changed pixels produce large standardized differences. The iterative weighting step reduces the influence of likely changed pixels while estimating the statistical no-change background.

IR-MAD is applied separately to the three-band optical red-green-blue (RGB) pair and the two-band SAR VV/VH pair. For each modality, it identifies paired combinations of before-event and after-event bands that behave similarly over stable terrain. The differences between those paired combinations form the MAD components. Each component is standardized using its weighted empirical variability so that components with different scales can be compared consistently. The standardized components are combined into a chi-square change measure, which is converted into a confidence value ranging from low confidence for stable pixels to high confidence for statistically unusual pixels [8]. Pixels that appear unchanged receive greater weight in the next iteration, allowing the background model to become progressively less influenced by actual changes. Iteration stops when the estimated no-change

TABLE I
SOURCE DATASETS AND ACQUISITION CHARACTERISTICS

Dataset	Spatial resolution	Temporal resolution	Acquisition window
BIPAD incident inventory [17]	Incident coordinate or reported zone; non-raster	Event based; records span 2018–2026	All collected landslide incident records.
Planet Global Quarterly Visual Mosaics [18]	4.77 m at the equator; red, green, and blue bands	Quarterly composites	Nearest published quarter before the incident and first available quarter at or after the incident.
Sentinel-1 ground-range detected SAR [19]	Approximately 10 m pixel spacing for the selected product	Approximately 6–12 days, depending on constellation coverage	Last available pre-event scene and first available post-event scene.
Sentinel-2 Level-2A surface reflectance [20]	10 m for visible and near-infrared bands; 20 m for the Scene Classification Layer (SCL)	Nominal five-day constellation revisit	Nearest cloud-acceptable pre-event scene when a suitable Planet pre-event mosaic is unavailable.
SRTM elevation [21]	One arc-second, approximately 30 m	Static mission acquisition from February 2000	One terrain observation for each incident zone.



(a) Planet optical image **(b)** Sentinel-1 SAR image **(c)** Terrain slope image

Fig. 2. Representative multi-source observations used by the workflow: (a) high-resolution Planet optical imagery, (b) Sentinel-1 SAR imagery, and (c) terrain slope.

probabilities stabilize. The process produces separate optical and SAR change-confidence maps, together with combined statistical evidence where both modalities are valid.

c) Confidence fusion and slope constraint: Confidence fusion combines complementary sensor evidence into one change-likelihood surface. Optical confidence is sensitive to reflectance and land-cover change, whereas SAR confidence responds to changes in surface roughness, structure, and moisture. The workflow uses a geometric mean, which behaves like an agreement rule: the fused confidence remains high only when both modalities provide substantial evidence. This reduces changes that are prominent in only one sensor. A minimum-slope mask is then applied as a terrain prior because landslide candidates are expected to occur on sufficiently inclined terrain; similar combinations of spectral change and terrain screening have been used in operational landslide-identification systems [2].

d) Simple Linear Iterative Clustering (SLIC) spatial regularization: A superpixel is a connected group of neighboring pixels with similar image characteristics. SLIC is an over-segmentation method that partitions an image into compact superpixels by balancing feature similarity and spatial proximity [9]. In this workflow, superpixels provide spatial regularization: isolated pixel-level responses are reduced, while groups of adjacent pixels that may belong to the same physical surface change are treated consistently.

The selected physical superpixel scale is converted to the common grid resolution. SLIC then groups nearby pixels according to both image similarity and spatial distance, with a compactness setting controlling whether segments follow detailed boundaries or form more regular shapes. The mean terrain-constrained confidence is assigned to each superpixel. Consequently, all pixels within a segment share a common response, suppressing isolated noise while retaining coherent changes that are more likely to represent physical objects [9].

e) Segment-level statistical significance: Pixel confidence alone can produce high responses by chance when many locations are examined. The workflow therefore sums the combined optical and SAR chi-square evidence within each superpixel and compares the accumulated evidence with the amount expected under a no-change model. Because many superpixels are tested within an incident, a Bonferroni correction adjusts the significance requirement according to the number of tested segments, controlling the probability of false detections across the incident [13]. Only statistically significant superpixels remain eligible. This gate allows an incident to produce no candidates when its apparent changes are consistent with noise.

f) Statistical hysteresis thresholding: Hysteresis thresholding uses two confidence levels rather than one. A strict

threshold identifies reliable change seeds, while a lower threshold admits weaker pixels only when they are connected to those seeds. This approach, originally established for preserving meaningful connected responses during edge detection, retains lower-confidence boundaries without accepting every weak response [14]. In this workflow, region growth uses horizontal, vertical, and diagonal neighbors and is limited to Bonferroni-significant superpixels that satisfy the slope constraint. Small residual objects are then removed using standard mathematical-morphology operations [15].

g) Component filtering and ranking: A connected component is a maximal group of foreground pixels linked by the selected neighborhood rule. Connected-component analysis converts the binary hysteresis mask into separate preliminary objects, following the standard region-labeling principle used in digital image processing [16]. Each object is assigned a physical area from its pixel count and ground resolution. Its elongation is estimated from the spread of its pixel coordinates along the major and minor axes. Objects outside the selected area and elongation limits are discarded. The remaining candidates are ranked using the product of their physical area and mean fused confidence, prioritizing regions that are both spatially substantial and strongly changed.

h) Candidate export and manual review: A candidate chip is a compact geospatial subset centered on one ranked component and containing the evidence needed for human interpretation. It reduces the review area while preserving local context around the detected change. Each ranked component is exported together with its corresponding before-event observation, after-event observation, terrain context, and change mask.

Manual review is the validation stage in which an analyst decides whether a candidate represents a landslide, a non-landslide change, or an uncertain case. Candidate image pairs are presented side by side through the QGIS annotation interface for classification, as illustrated in Fig. **??**. The plugin has been executed successfully, demonstrating that generated candidates can be loaded, compared, and reviewed within the manual annotation workflow. Human labels, rather than the automated change mask alone, determine which candidates are retained for segmentation dataset development.

C. Manual Review and Segmentation Annotation

Each generated candidate is examined in QGIS using the before-event image, after-event image, slope context, and candidate mask. The reviewer classifies the candidate as a landslide or non-landslide change. Confirmed landslides are then selected for pixel-level segmentation annotation so that the visible landslide extent, rather than only the candidate bounding region, can be used for supervised segmentation-model development.

The reviewed collection currently contains 248 landslide candidates and 5,595 non-landslide candidates. Segmentation masks have been completed for 164 of the confirmed landslides. Candidate classifications and segmentation annotations are retained with their incident identifiers and supporting imagery to preserve traceability between the original incident record, the detected change, and the final label.

IV. Quality Control and Reproducibility

Quality control is enforced through persistent processing records, deterministic incident identifiers, data-availability checks, and rerun-safe skip conditions. Successful operations are not repeated, while failed operations can be retried. Common-grid reprojection ensures that optical, SAR, and terrain observations are spatially aligned. Bonferroni correction limits incident-level false candidate generation, while area and elongation filters remove implausible objects. Manual candidate review prevents automated change masks from being treated directly as landslide labels, and segmentation annotations are linked to the corresponding confirmed candidates. Centralized storage preserves traceability from each reported incident to its observations, candidates, classifications, and segmentation masks.

V. Results and Dataset Status

Collection of the reported BIPAD landslide cases is complete. The image-acquisition, IR-MAD candidate-detection, QGIS review, and segmentation-annotation stages have been implemented. Candidate detection has been completed for the first 730 incidents. Manual review has identified 248 landslide candidates and 5,595 non-landslide candidates, for a total of 5,843 reviewed candidates. Pixel-level segmentation annotations have been completed for 164 landslides. Processing of the remaining incidents and continued annotation are in progress.

TABLE II
Current project data volumes

Artifact	Count
Reported incidents	4,138
Incidents processed for candidates	730
Total reviewed candidates	5,843
Landslide candidates	248
Non-landslide candidates	5,595
Segmentation-annotated landslides	164

The completed processing demonstrates that the workflow can move from incident records to reviewable, geospatially consistent candidate packages. The QGIS plugin successfully loads the generated evidence layers and records analyst decisions. The segmentation-annotation stage then converts confirmed landslide candidates into pixel-level masks suitable for supervised model development.

VI. Dataset Coverage Analysis

The coverage analysis used 4,138 reported incidents, 247 labeled landslides, and 5,594 non-landslide labels. The operational collection subsequently increased to the current totals in Table II, so the figures in this section remain one landslide and one non-landslide label behind the live dataset. The incident records span February 16, 2018 to May 20, 2026, whereas the labeled-landslide data used in the analysis span October 1, 2024 to August 11, 2025.

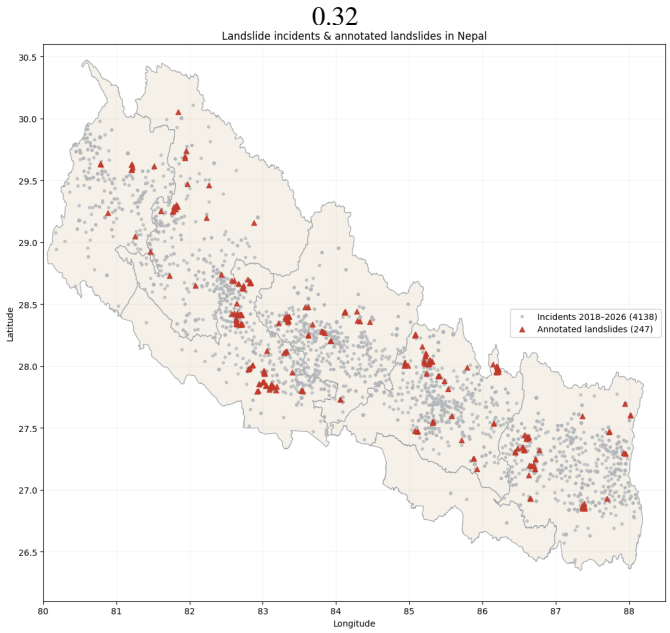


Fig. 3. Spatial distribution of 4,138 reported BIPAD landslide incidents and 247 labeled landslides across Nepal.

Geographic coverage extends across all seven provinces, as shown in Fig. 3. The analyzed labeled-landslide set contains 78 samples in Lumbini, 54 in Bagmati, 50 in Koshi, 31 in Gandaki, 22 in Karnali, 12 in Sudurpashchim, and none in Madhesh. All reported incidents and labeled landslides fall within the Nepal province polygons used in the analysis. Separately, 170 non-landslide labels fall outside those polygons, indicating a small subset that requires border or coordinate-quality review.

The elevation distributions show a lower central elevation for labeled landslides than for non-landslide labels. Labeled landslides have a mean elevation of 1,738.5 m and median of 1,655 m, compared with a mean of 2,598.1 m and median of 2,206.5 m for non-landslide labels. Figs. 4 and 5 also show a broader high-elevation tail among the non-landslide samples.

The dataset is strongly concentrated in the monsoon season. Of the 4,138 reported incidents, 3,524 (85.2%) occur during the monsoon, and 201 of the 247 labeled landslides (81.4%) fall in the same season. The monthly distributions are shown in Fig. 6. Physiographically, 59.1% of labeled landslides occur in the Middle Hills, followed by the Siwaliks (23.5%), High Mountains (16.2%), and High Himal (1.2%); no labeled landslide in the analyzed set falls within the Terai band. The physiographic representation is shown in Fig. 7. The reported incident zones are much larger than the exported review chips: their median area is approximately 37.2 km^2 and their median longest side is approximately 6.94 km, supporting the need for automated candidate localization before manual review.

VII. Discussion

The results show that incident-driven change detection can substantially reduce the area that must be inspected manually.

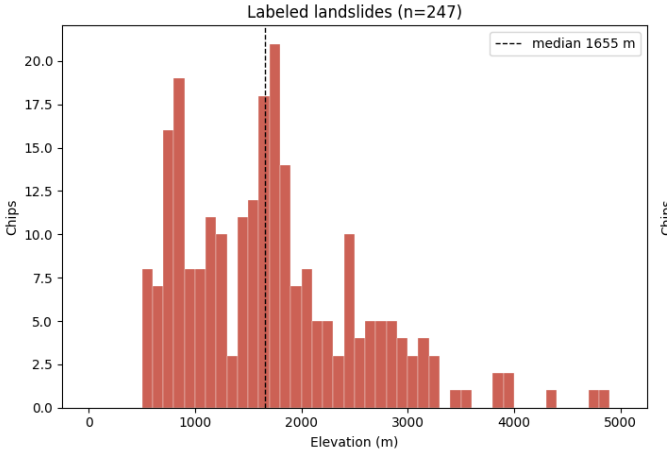


Fig. 4. Elevation distribution of labeled landslides. The dashed line indicates the median elevation.

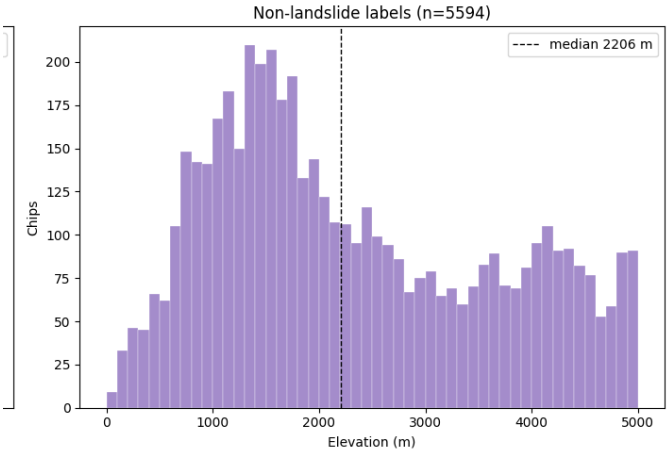
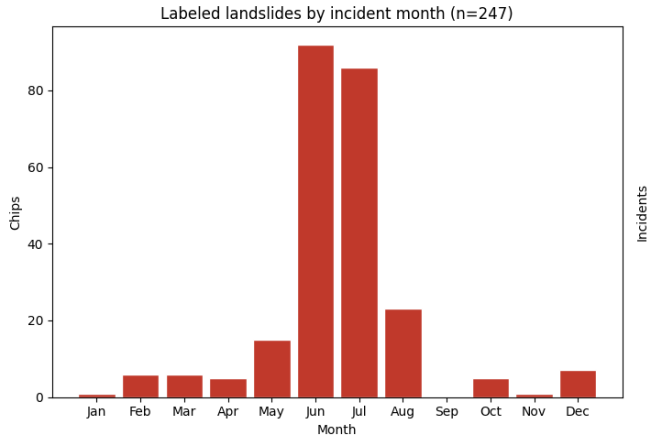


Fig. 5. Elevation distribution of non-landslide labels. The dashed line indicates the median elevation.

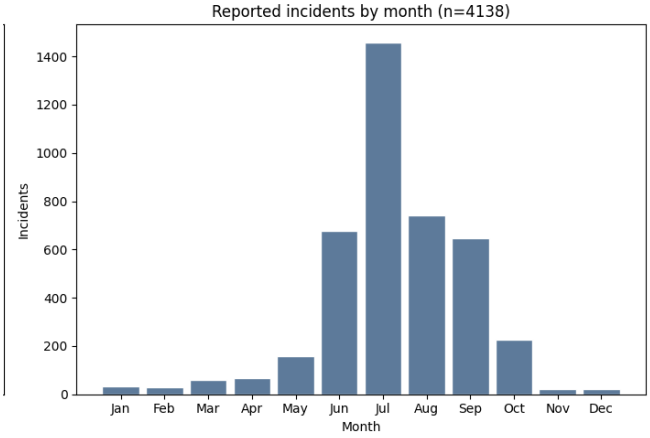
Instead of asking analysts to search complete incident zones, the workflow presents ranked regions supported by optical change, SAR change, terrain plausibility, and segment-level statistical evidence. This preserves human judgment while making review more focused and reproducible.

The reviewed data are strongly imbalanced: 248 of the 5,843 reviewed candidates are landslides, while 5,595 are non-landslide changes. This confirms that temporal change alone is not equivalent to landslide occurrence and highlights the value of retaining hard negative samples. Such negatives include changes caused by vegetation, agriculture, roads, rivers, illumination, moisture, and sensor effects, and they are important for training models that must distinguish landslides from visually similar disturbances.

The coverage analysis also reveals geographic, temporal, elevation, and physiographic imbalance. Labeled landslides are concentrated in the monsoon season and Middle Hills, with no labeled sample in Madhesh or the Terai band in the analyzed data. These imbalances should be considered when



(a) Labeled landslides



(b) Reported incidents

Fig. 6. Monthly distributions of (a) labeled landslides and (b) reported incidents.

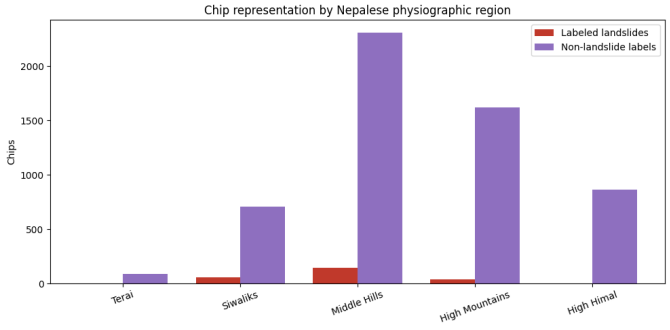


Fig. 7. Representation of labeled landslides and non-landslide labels across Nepal's physiographic regions.

splitting the data and interpreting model performance because a random split could overrepresent dominant regions while providing weak evidence of generalization to underrepresented environments.

The workflow contributes a continuously extensible dataset-construction process rather than a fixed benchmark. Its principal value is the traceable connection between a reported incident, the multi-source evidence used for candidate generation, the analyst's decision, and the final segmentation annotation. This structure supports later error analysis and enables the dataset to grow without discarding provenance.

VIII. Limitations and Future Work

BIPAD coordinates may describe affected settlements rather than exact failure locations, and quarterly Planet mosaics may smooth or temporally mix short-lived surface changes. Optical and SAR observations also respond differently to vegetation, moisture, shadow, layover, and land-cover change. Reprojection and resampling can introduce boundary uncertainty, while geometric fusion may suppress a true change that is strongly visible in only one modality. The chi-square aggregation further assumes an approximate statistical model that may not hold uniformly across terrain and acquisition conditions.

Only 730 of the 4,138 incidents have completed candidate detection, and manual labels remain class-imbalanced and subject to analyst interpretation. Segmentation annotations are available for 164 of the 248 confirmed landslide candidates, leaving part of the positive collection without pixel-level boundaries. The coverage analysis also identified 170 non-landslide labels outside the province polygons used for validation, requiring coordinate or boundary review. Future work will therefore process the remaining incidents, complete the outstanding segmentation masks, resolve questionable coordinates, measure inter-annotator agreement, compare detections with independent inventories, and evaluate baseline classification and segmentation models using province-aware, elevation-aware, and temporally separated validation data.

IX. Conclusion

This manuscript presents an implemented workflow that converts reported BIPAD incidents into reviewed landslide and non-landslide candidates and pixel-level segmentation annotations. Planet mosaics, Sentinel-1 SAR, Sentinel-2 support imagery, and SRTM terrain data are aligned and analyzed through IR-MAD. Geometric optical–SAR fusion, terrain filtering, SLIC aggregation, Bonferroni-corrected significance, hysteresis, morphology, and geometric screening produce ranked candidates for QGIS review. Candidate detection is complete for the first 730 incidents, producing 248 landslide candidates and 5,595 non-landslide candidates. Segmentation annotations have been completed for 164 landslides. The completed system establishes a reproducible foundation for processing the remaining incidents and expanding the segmentation dataset.

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